

Algebra II with Trigonometry

Chapter 6.3

Olympiad Problems (Gemini)

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[4531393_alg-2trig-h-chp-6-3-note.pdf](#)

Problems

1. Find the sum of all real solutions to the equation $x = 2024\pi \sin(x)$.
2. Determine the difference between the absolute maximum and absolute minimum values of the function $f(x) = (\sin x + \cos x)^4$ over the real numbers.
3. How many real numbers x satisfy the equation $\sin(x) + \cos(x\sqrt{2}) = 2$?
4. Find the sum of the absolute maximum and absolute minimum values of the function $f(x) = \cos(2x) - 4\sin(x)$.
5. The function $v(t) = \sin\left(\frac{t}{a}\right) + \cos\left(\frac{t}{b}\right)$, where a and b are relatively prime positive integers, models the combined sound wave of two tuning forks. If the fundamental period of $v(t)$ is 120π , find the minimum possible value of the sum $a + b$.
6. Determine the exact amplitude of the combined sinusoidal wave given by $y = \sin(x) + \sin\left(x + \frac{\pi}{3}\right) + \sin\left(x + \frac{2\pi}{3}\right)$.